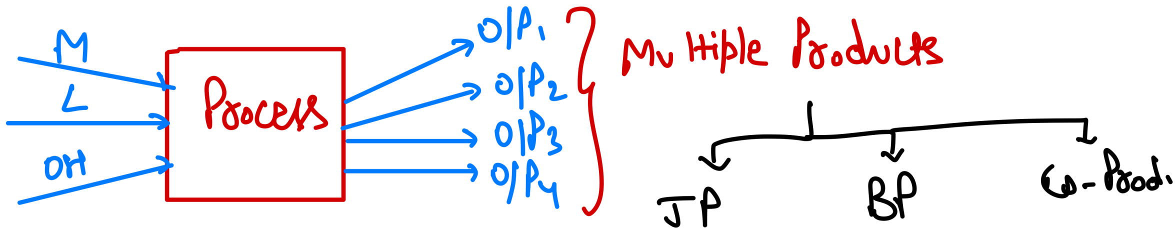
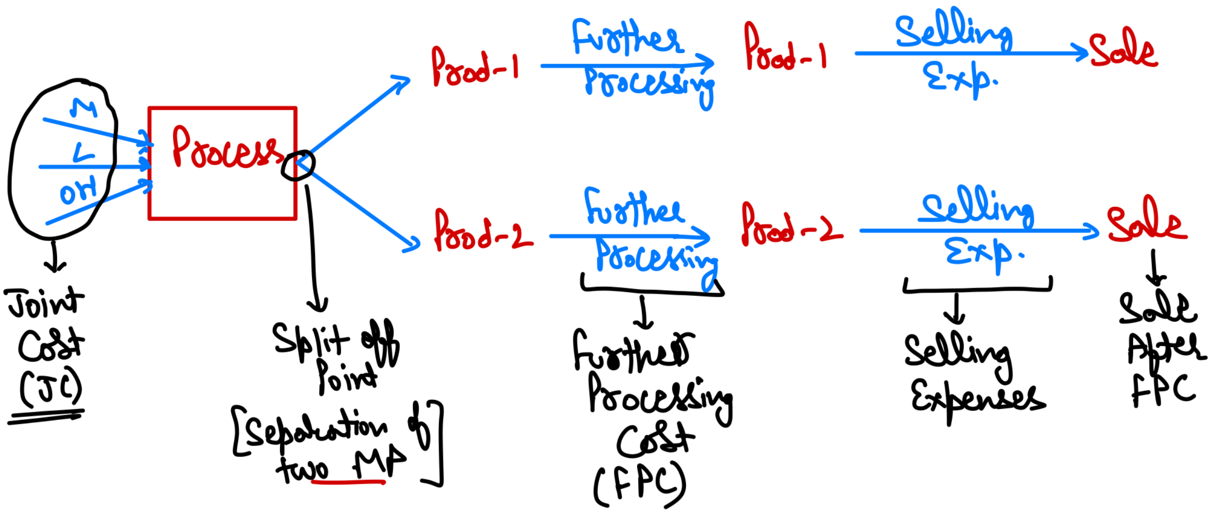
JOINT & BY-PRODUCT

1. Multiple Output



2. Meaning of Basic Terms



3. Treatment of By-Product

Treatment of BP

- BP of small value
  - Option – 1: Treat similar to Normal loss
  - Option – 2: Consider it as miscellaneous income and show in credit side of costing P&L A/c
- BP of considerable value or BP requires FPC
  - Treat as JP or MP i.e. calculate share of JC for this

4. Methods of Apportionment of Joint Cost (JC)

(A) Reverse Cost Method

Particulars	Product A	Product B
Sale value Less: Profit	- -	- -
Total Cost Less: Selling expenses	- -	- -
Work Cost Less: Further Processing cost	- -	- -
Share of Joint Cost		

If total joint cost is not matching with value given in question then distribute actual joint cost given in question in the share of joint cost calculated from above statement.

If data of profit is given for only one product then prepare above table for one product after that share of JC of other product = Total JC – Share of JC from above table for one product.

(B) Physical Unit Method – Distribute on the basis of physical units produced

(C) Sale value at split off method – Distribute on the basis of sale value at split off of units produced.

(D) Sale value after FPC method – Distribute on the basis of sale value after FPC of units produced.

(E) Net Realizable Value (NRV) Method – Distribute on the basis of NRV

NRV = Sale after FPC – Selling Expenses – FPC

(F) Contribution Margin Method

- Distribute variable cost on the basis of units produced
- Distributed fixed cost on the basis of contribution
- If any product has zero or negative contribution than FC will not be distributed for that product.

(G) Average Cost Method

$$\text{Average cost} = \frac{\textit{Total Joint Cost}}{\textit{Total Untis at Seperation Point}}$$

5. Decision Regarding Further Processing

Compare Incremental Revenue with Incremental Cost

If Incremental revenue > Incremental cost - Yes further process

If Incremental revenue < Incremental cost - Not to further process

Incremental revenue = Sale after FPC – Sale at split off

Incremental cost = FPC + Increase in any other cost – Decrease in any other cost

Question – 1

A company's plant processes 6,750 units of a raw material in a month to produce two products 'M' and 'N'. The process yield is as under:

Product M	80%
Product N	12%
Process loss	8%

The cost of raw material is ₹ 80 per unit.

Processing cost is ₹ 2,25,000 of which labour cost is accounted for 66%. Labour is chargeable to products 'M' and 'N' in the ratio of 100:80.

Prepare a comprehensive cost statement for each product showing:

(i) Apportionment of joint cost among products 'M' and 'N' and

(ii) Total cost of the products 'M' and 'N'.

Solution

Total joint cost = Raw material cost + Processing cost = (6,750 × 80) + 2,25,000 = ₹ 7,65,000

Total labour cost = 2,25,000 × 66% = ₹ 1,48,500

Joint cost other than labour cost = 7,65,000 – 1,48,500 = ₹ 6,16,500

Statement of Joint Cost Apportionment

Particulars	Product M	Product N
Labour Cost (1,48,500 in 100:80)	82,500	66,000
Cost other than labour cost (6,16,500 in 80:12)	5,36,087	80,413
Share of Joint Cost	6,18,587	1,46,413

Statement of Total Cost

Particulars	Product M	Product N
Raw material cost (5,40,000 in 80:12)	4,69,565	70,435
Labour Cost (1,48,500 in 100:80)	82,500	66,000
Other Processing Cost (76,500 in 80:12)	66,522	9,978
Share of Joint Cost	6,18,587	1,46,413

Question – 2

Three products X, Y and Z alongwith a byproduct B are obtained again in a crude state which require further processing at a cost of ₹5 for X; ₹4 for Y; and ₹2.50 for Z per unit before sale. The byproduct is however saleable as such to a nearby factory. The selling prices for the three main products and byproduct, assuming they should yield a net margin of 25% of cost, are fixed at ₹13.75, ₹8.75 and ₹7.50 and ₹1.00 respectively – all per unit quantity sold.

During a period, the joint input cost including the material cost was ₹90,800 and the respective outputs were:

X	8,000 units
Y	6,000 units
Z	4,000 units
B	1,000 units

By-product should be credited to joint cost and only the net joint costs are to be allocated to the main products. Calculate the joint cost per unit of each product and the margin available as a percentage on cost.

Solution

Computation of share of joint costs

Particulars	X (₹)	Y (₹)	Z (₹)
Selling price	13.75	8.75	7.50
Less: Profit @ 25% on cost or 20% on sales	2.75	1.75	1.50
Cost of sales	11.00	7.00	6.00
Less: Post split off cost	5.00	4.00	2.50
Joint cost per unit	6.00	3.00	3.50
Output (units)	8,000	6,000	4,000
Total cost	48,000	18,000	14,000
Actual share of net joint cost (A) (90,000 in 48:18:14)	54,000	20,250	15,750
Output (units) (B)	8,000	6,000	4,000
Joint cost per unit (A ÷ B)	6.75	3.38	3.94

Statement of Profit

Particulars	X	Y	Z
Joint cost per unit	6.75	3.38	3.94
Post split off cost	5.00	4.00	2.50
Total cost	11.75	7.38	6.44
Selling price	13.75	8.75	7.50
Margin	2.00	1.37	1.06
Margin % on cost	17.02%	18.56%	16.46%

Working Note:

1) Joint input cost including material cost		90,800
Less: Realization from by-product		
Sale value (1,000 × 1)	1,000	
Less: Profit (1,000 × 20%)	200	800
Net Joint costs to be allocated		90,000

Question – 3

A factory producing article P also produces a by-product Q which is further processed into finished product. The joint costs of manufacture are given below:

	₹
Material	5,000
Labour	3,000
Overheads	2,000
	10,000

Subsequent costs are given below:

	P	Q
Material	₹ 3,000	₹ 1,500
Labour	1,400	1,000
Overheads	600	500
	5,000	3,000

Selling prices are: P – ₹ 16,000 and Q – ₹ 8,000

Estimated profits margin on selling prices are 25% for P and 20% for Q.

Assume that selling and distribution expenses are in proportion of sales prices. Show how you would apportion joint costs of manufacture and prepare a statement showing cost of production of P and Q.

Solution

Statement of Cost

Particulars	Product P	Product Q
Sales	16,000	8,000
Less: Profit	16,000×25% = 4,000	8,000×20% = 1,600
Total Cost	12,000	6,400
Less: Selling expenses (w.n. – 1)	267	133
Work Cost	11,733	6,267
Less: Further Processing Cost	5,000	3,000
Share of Joint Cost	6,733	3,267

Working Note-1

Total sales = Total Joint Cost + Total Further Processing Cost + Total Selling Expenses + Total Profit

24,000 = 10,000 + 8,000 + Total Selling Expenses + 5,600

Total Selling Expenses = ₹ 400

Ratio of sales = 16,000 : 8,000 = 2:1

Selling expenses of P = 400 × (2/3) = ₹ 267

Selling expenses of Q = 400 × (1/3) = ₹ 133

Statement of Cost of Production

Particulars	Product P	Product Q
Joint Cost:		
Material (5,000 in 6,733:3,267)	3,367	1,633
Labour (3,000 in 6,733:3,267)	2,020	980
Overheads (2,000 in 6,733:3,267)	1,347	653
Further Processing Cost:		
Material	3,000	1,500
Labour	1,400	1,000
Overheads	600	500
Cost of Production	11,733	6,267

Question – 4

The SK Oil company purchase crude vegetable oil. It does refining of the same. The refining process results in four products at the split off point: M, N, O & P.

Product O is fully processed at the split off point. Product M, N, & P can be individually further refined into 'Super M', 'Super N' and 'Super P'. In the most recent month (October), the output at split off point was:

Product M	3,00,000 gallons
Product N	1,00,000 gallons
Product O	50,000 gallons
Product P	50,000 gallons

The joint cost of the purchasing the crude vegetable oil and Processing it were ₹ 40,00,000. SK had no beginning or ending inventories. Sales of Product O in October were ₹ 20,00,000. Total output of products M, N and P was further refined and then sold. Data related to October are as follows:

Product	***Further processing cost to make super products***	***Sales***
Super M	₹ 80,00,000	₹ 1,20,00,000
Super N	₹ 32,00,000	₹ 40,00,000
Super P	₹ 36,00,000	₹ 48,00,000

SK had the option of selling products M, N and P at the split off point. This alternative would have yielded the following sales for the October, production:

Product M	₹ 20,00,000
Product N	₹ 12,00,000
Product P	₹ 28,00,000

You are required to answer:

(a) How the joint cost of ₹ 40,00,000 would be allocated between each product under the following methods

(i) Sales value at split off point

(ii) Physical output (gallon)

(iii) Estimated net realizable value

(b) Could SK have increased its October operating profits by making different decisions about the further refining of product M, N or P? Show the effect of any change you recommend on operating profits.

Solution

(a) (i) Statement of Cost (₹ in lakhs)

Particulars	**Product M**	**Product N**	**Product O**	**Product P**
Sale value at split off	20	12	20	28
Share of Joint Cost (40 in 20:12:20:28)	10	6	10	14

(a) (ii) Statement of Cost (₹ in lakhs)

Particulars	**Product M**	**Product N**	**Product O**	**Product P**
Physical Output	3	1	0.50	0.50
Share of Joint Cost (40 in 3:1:0.5:0.5)	24	8	4	4

(a) (iii) Statement of Cost (₹ in lakhs)

Particulars	**Product M**	**Product N**	**Product O**	**Product P**
Sale value after FPC	120	40	20	48
Less: Further Processing cost	80	32	-	36
Less: Selling expenses	-	-	-	-
Net Realizable Value	40	8	20	12
Share of Joint Cost (40 in 40:8:20:12)	20	4	10	6

(b) Statement of Incremental Profit/(Loss) (₹ in lakhs)

Particulars	**Product M**	**Product N**	**Product P**
Sales after FPC	120	40	48
Less: Sale at Split off	20	12	28
Incremental Sales	100	28	20
Less: Further Processing Cost	80	32	36
Incremental Profit/(Loss)	20	(4)	(16)

Thus, it is recommended that Product M should be further processed into Super M and Product N & P should be sold at split off point without any further processing. With this suggestion, company's operating profit will rise by ₹ 20,00,000 as the existing losses of ₹ 4,00,000 and ₹ 16,00,000 will get cut down.

Question – 5

SK Ltd. is engaged in the production of Buttermilk, Butter and Ghee. It purchases processed cream and let it through the process of churning until it separates into buttermilk and butter. For the month of May, 2021, SK Ltd. purchased 50 kilolitre processed cream @ ₹ 100 per 1,000 ml. Conversion cost of ₹ 1,00,000 were incurred up-to the split off point, where tow saleable products were produced i.e. buttermilk and butter. Butter can be further processed into Ghee.

The May, 2021 production and sales information is as follows:

Products	Production (in Kilolitre/tonne)	Sales Quantity (in Kilootire/tonne)	Selling price per litre/kg (₹)
Buttermilk	28	28	30
Butter	20	-	-
Ghee	16	16	480

All 20 tonne of butter were further processed at an incremental cost of ₹ 1,20,000 to yield 16 Kilolitre of Ghee. There was no opening or closing inventories of buttermilk, butter or ghee in May, 2021.

Required:

(i) Show how joint cost would be apportioned between Buttermilk and Butter under estimated net realizable value method.

(ii) MP Ltd. offers to purchase 20 tonne of butter in June at ₹ 360 per kg. In case SK Ltd. accepts this offer, no Ghee would be produced in June. Suggest whether SK Ltd. shall accept the offer affecting its operating income or further process butter to make Ghee itself?

Solution

(i) Total Joint Cost = Processed cream cost + conversion cost

$$= (50 \times 1,000 \times 100) + 1,00,000 = ₹\ 51,00,000$$

Statement of Joint Cost

Particulars	Buttermilk Amount (₹)	Butter Amount (₹)
Sales Value	30×28×1,000 = 8,40,000	16×1,000×480 = 76,80,000
Less: Post split-off cost	-	(1,20,000)
Net Realizable Value	8,40,000	75,60,000
Apportionment of Joint Cost of ₹ 51,00,000 in ratio of 1:9	5,10,000	45,90,000

(ii) Statement of Incremental Profit or Loss

Particulars	**(₹)**
Revenue from Ghee	16 × 1,000 ×480 = 76,80,000
(-) Revenue from Butter	20 × 1,000 × 360 = 72,00,000
Incremental Revenue	4,80,000
(-) Further processing cost	1,20,000
Incremental Profit	3,60,000

The operating income of SK Ltd. will be reduced by ₹ 3,60,000 in June if it sells 20 tonne of Butter to MP Ltd., instead of further processing of Butter into Ghee for sale. Thus, SK Ltd. is advised not to accept the offer and further process butter to make Ghee itself.

Question – 6

A company produces two products A and B, through a joint production process. The total joint production cost incurred is as under:

Material	₹ 20,000
Labour	₹ 10,000
Variable overheads	₹ 6,000
Fixed overheads	₹ 24,000

Product A and B can be sold for ₹ 20 per unit and ₹ 15 per unit respectively at split off point. The produced quantities are Product A-2,000 units and Product B – 4,000 units.

(i) You are required to calculate the joint production cost allocation for each product using the:
   (a) Physical unit method.
   (b) Contribution margin method.

(ii) Product B can be further processed by incurring expenditure of ₹ 12,000. Loss in further processing is 2%. It can be sold @ ₹ 18 per unit. Explain the impact on profitability if Product B is further processed.

Solution

(i) (a) Total Joint Cost = 20,000 + 10,000 + 6,000 + 24,000 = ₹ 60,000

Share of Joint cost of Product A = $60,000 \times \frac{2,000}{6,000}$ = ₹ 20,000

Share of Joint cost of Product B = $60,000 \times \frac{4,000}{6,000}$ = ₹ 40,000

(b) Total Variable cost = 20,000 + 10,000 + 6,000 = ₹ 36,000

Total Fixed cost = ₹ 24,000

Statement of Cost

Particulars	**Product A**	**Product B**
Sales Value (A)	2,000 × 20 = 40,000	4,000 × 15 = 60,000
Variable Cost (B) [36,000 in quantity ratio 2:4]	12,000	24,000

Contribution (A – B)	28,000	36,000
Fixed cost (C) [24,000 in 28:36]	10,500	13,500
Share of Joint Cost (B + C)	22,500	37,500

(ii) **Statement of Profit of Product B**

Particulars	Amount
Units produced and sold 98% of 4,000 units Selling price per unit	3,920 18
Sales value after FPC Sales value at split off	70,560 60,000
Incremental revenue Less: Further processing cost	10,560 12,000
Incremental loss after further processing	(1,440)

Thus, there is a net loss of ₹ 1,440 if the Product B is further processed and sold.

Question – 7

ABC Company produces a Product 'X' that passes through three processes: R, S and T. Three types of raw materials, viz., J, K and L are used in the ratio of 40:40:20 in process R. The output of each process is transferred to next process. Process loss is 10% of total input in each process. At the stage of output in process T, a by-product 'Z' is emerging and the ratio of the main product 'X' to the by-product 'Z' is 80:20. The selling price of product 'X' is ₹ 60 per kg.

The company produced 14,580 kgs of product 'X'.
Material price: Material K @ ₹ 15 per kg; Material K @ ₹ 9per kg.
Material L @ ₹ 7 per kg. Process costs are as follows:

Process	Variable cost per kg (₹)	Fixed cost of input (₹)
R	5.00	42,000
S	4.50	5,000
T	3.40	4,800

The by-product 'Z' cannot be processed further and can be sold at ₹ 30 per kg at the split-off stage. There is no realizable value of process losses at any stage.

Required:
Present a statement showing the apportionment of joint costs on the basis of the sales value of product 'X' and by-product 'Z' at the split-off point and the profitability of product 'Z' and by-product 'Z'.

Solution

Working Note:

(1) Let total raw material in Process R of raw material R be 100

Process	Input	Loss	Output
R	100	$100 \times 10\% = 10$	90
S	90	$90 \times 10\% = 9$	81
T	81	$81 \times 10\% = 8.10$	72.90

Thus, for input of 100 units in process R, output of 72.90 units is obtained from process T.

Actual output of X = 14,580 units

80% of output of Process T = 14,580 units

Output of process T = 14,580 ÷ 80% = 18,225

Input of process R = $\frac{18,225}{72.9} \times 100$ = 25,000 kgs

(2) Calculation of Joint cost

Process	Inputs	Variable costs	Fixed cost	Total cost
R	25,000	$25,000 \times 5 = 1,25,000$	42,000	1,67,000
S	22,500	$22,500 \times 4.5 = 1,01,250$	5,000	1,06,250
T	20,250	$20,250 \times 3.4 = 68,850$	4,800	73,650
				3,46,900

Raw material J	= 10,000 × 15	=	₹ 1,50,000
Raw material K	= 10,000 × 9	=	₹ 90,000
Raw material L	= 5,000 × 7	=	₹ 35,000
			₹ 2,75,000
Add: Processing cost (as above)			₹ 3,46,900
Total joint cost			₹ 6,21,900

(i) Statement showing apportionment of joint cost

Particulars	Product X	By-Product Z	Total
Units	14,580	3,465	
Selling price (₹)	60	30	
Sales value (₹)	8,74,800	1,09,350	9,84,150
Share of joint cost (₹ 6,21,900 in ratio of sales value)	5,52,800	69,100	6,21,900

(ii) Statement of profitability

Particulars	Product X	By-Product Z	Total
Sales value (₹)	8,74,800	1,09,350	9,84,150
(-) Share of joint cost	(5,52,800)	(69,100)	(6,21,900)
Profit	3,22,000	40,250	3,62,250

Question – 8

Mayura Chemicals Ltd. buys a particular raw material at ₹ 8 per litre. At the end of the processing in Department-1, this raw material splits-off into products X, Y and Z. Product X is sold at the split-off point, with no further processing. Products Y and Z require further processing before they can be sold. Product Y is processed in Department-2, and Product Z is processed in Department-3. Following is a summary of the costs and other related data for the year 2019-20:

Particulars	Department		
	1	2	3
Cost of Raw Material	₹ 4,80,000	-	-
Direct Labour	₹ 70,000	₹ 4,50,000	₹ 6,50,000
Manufacturing Overhead	₹ 48,000	₹ 2,10,000	₹ 4,50,000
	Products		
	X	**Y**	**Z**
Sales (litres)	10,000	15,000	22,500
Closing inventory (litres)	5,000	-	7,500
Sale price per litre (₹)	30	64	50

There were no opening and closing inventories of basic raw materials at the beginning as well as at the end of the year. All finished goods inventory in litres was complete as to processing. The company uses the Net-realizable value method of allocating join costs.

You are required to prepare:

(i) Schedule showing the allocation of joint costs

(ii) Calculate the cost of goods sold of each product and the cost of each item in Inventory

(iii) A comparative statement of Gross Profit

Solution

(i) Statement of allocation of joint cost

Particulars	Product X	Product Y	Product Z	Total
Units sold	10,000	15,000	22,500	
Add: Closing stock (A)	5,000	-	7,500	
Units Produced (B)	15,000	15,000	30,000	
Selling price per unit (C)	30	64	50	
Sale value of Prod. (B × C)	4,50,000	9,60,000	15,00,000	29,10,000
Less: Additional cost	-	6,60,000	11,00,000	17,60,000
Net realizable value	4,50,000	3,00,000	4,00,000	11,50,000
Share of joint cost (D) (5,98,000 in NRV ratio)	2,34,000	1,56,000	2,08,000	5,98,000

(ii) **Statement of calculation of cost of goods sold and inventory**

Particulars	Product X	Product Y	Product Z	Total
Share of joint cost	2,34,000	1,56,000	2,08,000	5,98,000
Add: Additional costs	-	6,60,000	11,00,000	17,60,000
Less: Cost of inventories	$\frac{2,34,000 \times 5,000}{15,000} =$ (78,000)	-	$\frac{13,08,000 \times 7,500}{30,000}$ = (3,27,000)	(4,05,000)
Cost of goods sold	1,56,000	8,16,000	9,81,000	19,53,000

(iii) **Statement of calculation of gross profit**

Particulars	Product X	Product Y	Product Z	Total
Units sold	10,000	15,000	22,500	
Selling price per unit	30	64	50	
Sales	3,00,000	9,60,000	11,25,000	23,85,000
Less: Cost of goods sold	1,56,000	8,16,000	9,81,000	19,53,000
Profit / (loss)	1,44,000	1,44,000	1,44,000	4,32,000

Question – 9

A Company produces two joint products P and Q in 70:30 ratio from basic raw materials in department A. The input output ratio of department A is 100:85. Product P can be sold at the split off stage or can be processed further at department B and sold as product AR. The input output ratio is 100:90 of department B. The department B is created to process product P only and to make it product AR.

The selling prices per kg are as under:

Product P ₹ 85

Product Q ₹ 290

Product AR ₹ 115

The production will be taken up in the next month.

Raw materials 8,00,000 kgs

Purchase price ₹ 80 per kg

	Deptt. A (₹ Lacs)	Deptt. B (₹ Lacs)
Direct materials	35.00	5.00
Direct labour	30.00	9.00
Variable overhead	45.00	18.00
Fixed overheads	40.00	32.00
Total	150.00	64.00
Selling Expenses	**₹ in lacs**	
Product P	24.60	
Product Q	21.60	
Product AR	16.80	

Required:

(i) Prepare a statement showing the apportionment of joint costs if Joint Costs are apportioned in the proportion of NRV at Split-off point.

(ii) State whether it is advisable to produce product AR or not.

Solution

(a) Input in Department A = 8,00,000 units

Output in Department A = 8,00,000 × 85% = 6,80,000 units

Output of Product P = 6,80,000 × (70/100) = 4,76,000 units

Output of Product Q = 6,80,000 × (30/100) = 2,04,000 units

Statement of Joint Cost

Particulars	**Amount (₹ in lakhs)**
Raw Material (8,00,000 × 80)	640
Direct materials	35
Direct labour	30
Variable overheads	45
Fixed overheads	40
Total Joint Cost	790

Statement of Apportionment of Joint Cost **(₹ in lakhs)**

Particulars	**Product P**	**Product Q**
Sale at split off	4.76 × 85 = 404.60	2.04 × 290 = 591.60
Less: Selling expenses	24.60	21.60
Net realizable value at split off	380	570
Share of Joint Cost (790 in 38:57)	316	474

(b) Statement of Evaluation of Proposal

Particulars	**Amount (₹ in lakhs)**
Sales after further processing cost (w.n.-1)	492.66
(-) Sale at split off	404.60
Incremental sales	88.06
(-) Further processing cost	64
(+) Savings in selling expenses (24.60 – 16.80)	7.80
Net Benefit	31.86

Since there is net benefit, thus it is recommended to further process Product P into Product AR.

Working note – 1

Input in Department B = 4,76,000 units

Output in Department B i.e. Product AR = 4,76,000 × 90% = 4,28,400 units

Sale value after further processing cost = 115 × 4.284 lakhs = ₹ 492.66 lakhs

Question – 10

eSalt is the biggest producer of sodium hydroxide in India. This main product of the company has a strong reactivity with other organic compounds. It is highly versatile and is alkaline in nature. However, the basic material required for the production of this product is salt along with the electricity.

The manufacturing process involve electrolysis which produces Halogen as co-product. Modern use of Halogen is widespread. However, the common use is in disinfection like for purifying drinking water or swimming pool water. It is also an important ingredient of toothpaste. Thus, the company's management affirmed the simultaneous production of Halogen.

During the previous financial year, the company purchased the base material of ₹ 5,34,000. For the current year, company decided to increase the production by 2 times. Due to increased production, the total conversion cost hiked to 3 times. Last year, the conversion cost accounted to ₹ 8,01,000 up to the point at which two products i.e. sodium hydroxide and Halogen are separated.

The production and sales information for current year is provided as below:

	Sodium hydroxide	Halogen
Production/ Sales (in tonne)	24,030	16,020
Selling price per tonne (₹)	100	150

During the current year, the management of the company pointed the extensive use of Vinly which can be produced by further processing Halogen. Having selling price of ₹ 250 per tonne higher than that of the Halogen, it was decided not to sell Halogen and further process it into Vinly. The incremental processing cost took ₹ 8,01,000 producing 10,012.50 tonnes of Vinyl.

You are required to figure out the following for managerial decision:

Question – 1

For the current year, the amount of base material purchased and the conversion cost up to the point at which two products i.e. Sodium hydroxide and Halogen are separated would be:

A. base material ₹ 10,68,000 and conversion cost ₹ 24,03,000
B. base material ₹ 10,68,000 and conversion cost ₹ 16,02,000
C. base material ₹ 16,02,000 and conversion cost ₹ 24,03,000
D. base material ₹ 24,03,000 and conversion cost ₹ 16,02,000

Question – 2

Joint cost to be apportioned between Sodium hydroxide and Halogen as per the physical unit method would be:

A. Sodium hydroxide ₹ 24,03,000 and Halogen ₹ 10,68,000
B. Sodium hydroxide ₹ 10,68,000 and Halogen ₹ 16,02,000
C. Sodium hydroxide ₹ 16,02,000 and Halogen ₹ 24,03,000
D. Sodium hydroxide ₹ 24,03,000 and Halogen ₹ 16,02,000

Question – 3

Joint cost to be apportioned between Sodium hydroxide and Halogen as per the sales value at split- off point method would be:

A. Sodium hydroxide ₹ 20,02,500 and Halogen ₹ 20,02,500

B. Sodium hydroxide ₹ 16,02,000 and Halogen ₹ 24,03,000

C. Sodium hydroxide ₹ 24,03,000 and Halogen ₹ 16,02,000

D. Sodium hydroxide ₹ 10,68,000 and Halogen ₹ 20,02,500

Question – 4

Joint cost to be apportioned between Sodium hydroxide and Halogen as per the estimated net realisable value method would be:

A. Sodium hydroxide ₹ 23,44,390 and Halogen ₹ 16,60,610

B. Sodium hydroxide ₹ 17,16,429 and Halogen ₹ 22,88,571

C. Sodium hydroxide ₹ 22,88,571 and Halogen ₹ 17,16,429

D. Sodium hydroxide ₹ 16,60,610 and Halogen ₹ 23,44,390

Question – 5

Considering that the decision relating to further processing Halogen is not approved, suggest whether this would be in favour of the management by calculating incremental revenue/loss from further processing Halogen into Vinyl.

A. Incremental loss would be ₹ 16,02,000, thus the decision of not further processing Halogen is correct.

B. Incremental loss would be ₹ 8,01,000, thus the decision of not further processing Halogen is correct.

C. Incremental revenue would be ₹ 8,01,000, thus the decision relating to further processing Halogen needs to be approved.

D. Incremental revenue would be ₹ 16,02,000, thus the decision relating to further processing Halogen needs to be approved.

1	2	3	4	5
C	D	A	B	C